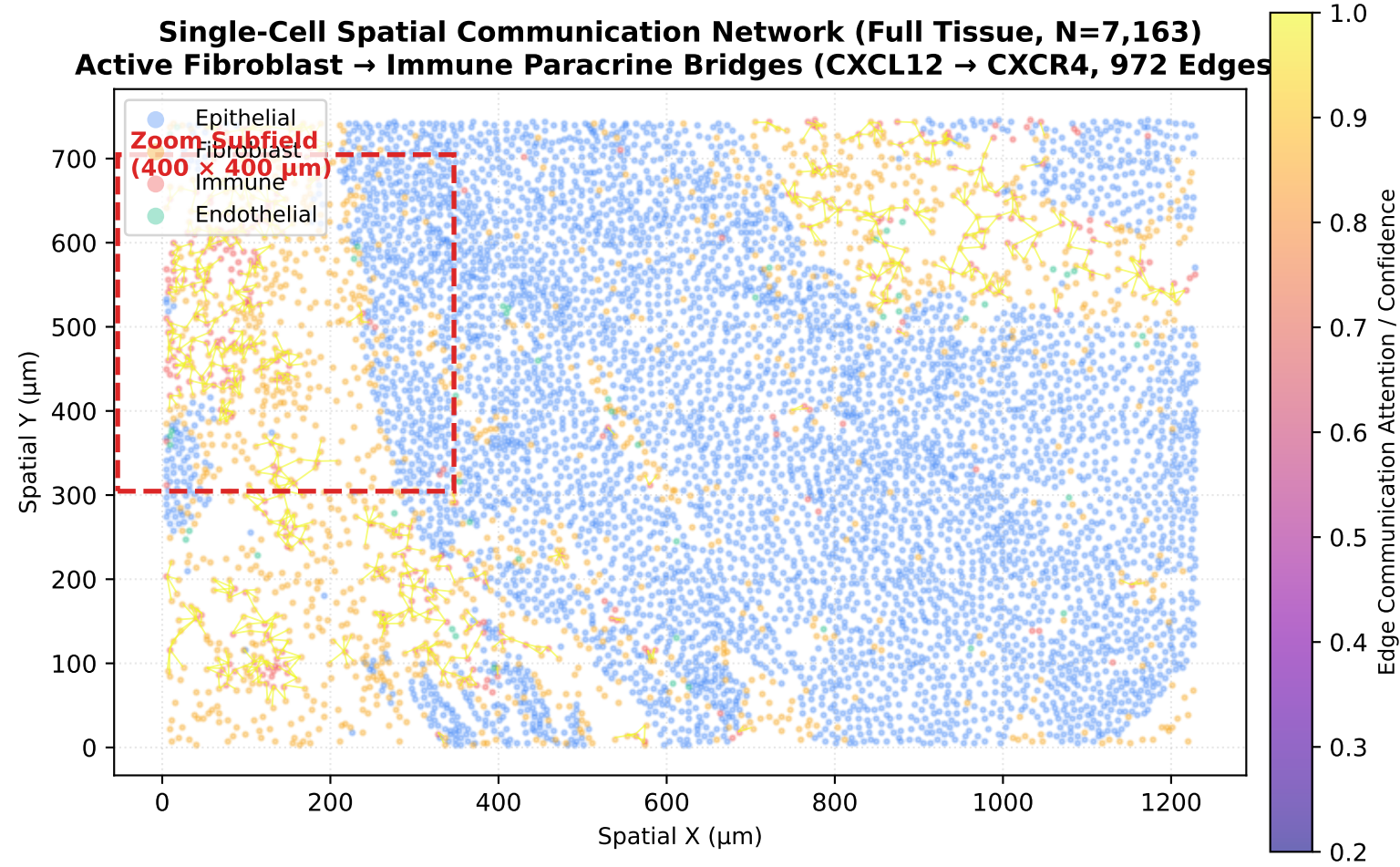


Spatial-Centric Single-Cell Communication Networks: In Situ CXCL12-CXCR4 Paracrine Signaling Architecture

Single-Cell Spatial Communication Network (Full Tissue, N=7,163)
Active Fibroblast → Immune Paracrine Bridges (CXCL12 → CXCR4, 972 Edges)



Invasive Margin Micro-Domain Zoom (400 × 400 μm)
Direct Paracrine Signaling Vectors (Fibroblast → Immune Cells)

